

***“An investigation into energy performance across
commercial buildings using predictive analysis models”***

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Declaration

I, Harry Byrne, declare that the thesis entitled “An investigation into energy performance across commercial buildings using predictive analysis models” was carried out by me for the degree of B.Sc. Business Analytics under the guidance and supervision of Neil O’Connor, Technological University Dublin. The interpretations put forth are based on my reading and understanding of the original texts and they are not published anywhere in the form of books, monographs or articles. The books, articles and websites that I have made use of are acknowledged at the respective place in the text. This thesis is my own original work and contains no material that has been previously submitted, in whole or in part, for the award of any other academic degree.

Signed: *Harry Byrne*

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Table of Contents

Acknowledgements.....	3
Table of Contents.....	4
Table of Figures.....	6
Chapter One: Introduction	8
1.1 Background.....	8
1.2 Research question	9
1.3 Research objectives.....	9
1.4 Structure of the Study	9
1.5 Environmental Impact:.....	10
Chapter 2: Literature Review	11
2.1 The structure of the literature review.....	11
2.2 Articles on the models used with similar building and energy data.....	11
2.3 Articles that look at different approaches to energy data.....	13
2.4 Discussion.....	16
2.5 Similarities Across Approaches	16
2.5.1 Machine Learning and Predictive Analytics	16
2.5.2 Energy Benchmarking and Urban Analysis.....	16
2.5.3 Sustainability and Energy Optimization	17
2.6 Differences in Methodologies.....	17
2.6.1 Use of Real-World vs. Synthetic Datasets	17
2.6.2 Building Type and Scope of Analysis.....	18
2.6.3 Role of External Variables and Climate Considerations.....	18
2.7 Patterns across the Literature Review:.....	18
2.7.1 Common Machine Learning Techniques:.....	18
2.7.2 Feature Engineering:.....	19
2.7.3 The integration of Environmental and Sustainability Goals:	19
2.7.4 Real-World data vs Synthetic data:.....	20
2.7.5 Emphasis on Explainability:	20
2.8 Emerging trends in Predictive Energy Modelling.....	20
Chapter 3: Design	22
3.1 Research Methodology	22
3.1.1 Business Understanding.....	22
3.1.2 Data Understanding.....	22
3.1.3 Data Preparation.....	23
3.1.4 Modelling.....	23

3.1.5 Evaluation	23
3.1.6 Deployment.....	23
3.2 Data Source.....	23
3.3 Data Preparation.....	25
3.3.1 Handling Missing Values.....	25
3.3.2 Encoding Categorical Variables.....	25
3.3.3 Feature Engineering	25
3.3.4 Outlier Detection.....	25
3.3.5 Scaling.....	26
3.4 Exploratory Data Analysis (EDA):	27
3.5 Modeling Approaches:.....	29
3.5.1 Linear Regression and Ridge Regression	29
3.5.2 Support Vector Regression:	30
3.5.3 Random Forest Regression:	30
3.5.4 XGBoost Regression:.....	30
3.5.5 Stacking Ensemble:.....	30
3.6 Model Evaluation:.....	30
3.7 Model Explainability with SHAP:	31
Chapter 4: Analysis.....	32
4. 1 Data Preparation.....	32
4.2 Exploratory Data Analysis (EDA)	32
4.3 Distribution of Energy Star Scores	33
4.4 Outliers.....	33
4.5 Model Training and Evaluation:	34
4.5.1 Baseline Models:.....	34
4.5.2 Support Vector Regression:	35
4.5.3 Tree Based Models:	35
4.5.4 Stacking Ensemble:.....	36
4.6 Hyperparameter Tuning and Model Optimisation:	36
4.6.1 Random Forest Tuning:	37
4.6.2 XGBoost Tuning:.....	37
4.6.3 The Importance of Hyperparameter Tuning:	37
4.7 Model Explainability with SHAP:	38
4.8 Overall Evaluation and Comparisons:	40
Figure 15 shows the results from this study compared to the results across other various studies.....	40
4.8.1 Ridge Regression:	40

4.8.2 Random Forest:	41
4.8.3 Support Vector Regression (SVR):	41
4.8.4 XGBoost Regressor:	41
4.8.5 Ensemble Stacking Model:	41
4.9 Classification Analysis:	42
4.10 Summary:	43
Chapter 5: Conclusions	44
5.1 Discussion of Findings.....	44
5.1.1 Feature Importance and Explainability	44
5.1.2 Benchmarking against existing Literature	44
5.1.3 Environmental and Policy Implications	45
5.2 Limitations of the analysis	45
5.2.1 Data Quality and Granularity	45
5.2.2 Unbalanced Feature Representation.....	45
5.2.3 Simplification of Environmental Factors	46
5.2.4 Model Interpretability vs Complexity trade off	46
5.2.5 General area of Focus	46
5.3 Model Bias and Overfitting.....	46
5.3.1 Risk of Overfitting	46
5.3.2 Risk of Bias.....	47
5.3.3 Addressing these Risks	47
5.4 Areas of Future Research.....	47
5.4.1 Incorporating Time-Series and Smart Meter Data	47
5.4.2 Extending to other countries	48
5.4.3 Integration of Geospatial Data	48
5.4.4 Environmental Lifecycle Modeling	49
5.4.5 Optimising Cost and Carbon factors	49
5.4.6 Real Time Energy Feedback	49
5.4.7 Policy Modeling.....	49
5.5 Conclusion	49

Table of Figures

Figure 1. Energy Usage Intensity.....	8
Figure 2. Analysis Framework.....	10
Figure 3. Energy Star Score by Property Type	27
Figure 4. Distribution of the top 10 building types.....	29
Figure 5. Top 10 Features correlated with Energy Star Score	32

Figure 6. Distribution of Energy Star Score	33
Figure 7. Outliers	34
Figure 8. Linear Regression Results	35
Figure 9. Ridge Regression Results	35
Figure 10. Support Vector Regression Results	35
Figure 11. Random Forest Results	35
Figure 12. XGBoost Results	36
Figure 13. Stacking Ensemble Results.....	36
Figure 14. Every Models result.....	36
Figure 15. Model Results Comparison	38
Figure 16. SHAP	39
Figure 17. A table of all the results compared to other studies	40
Figure 18. ROC Curve Graph	42
Figure 19. Confusion Matrix showing the classification results.....	43
Figure 20. Geographic Clustering of Buildings	48

Abstract

The urgency of climate change has made building energy efficiency an increasingly important factor in urban sustainability efforts. This study aimed to Predict energy Star Scores using buildings energy consumptions, and the building characteristics while also keeping with the sustainability goals set out. As buildings are among the largest contributors to energy consumption and carbon emission, rating systems have been developed, one of the key benchmarking rating systems in America is the Energy Star Score which indicates on a range from 1 -100 to show how well a building performs relative to similar buildings nationwide. This study explores the application of machine learning models, data exploration, clustering, and feature engineering to predict Energy Star Scores for commercial buildings in Chicago, using a publicly available dataset on energy benchmarking. Accurately predicting these scores is important as it supports energy management, identifies inefficient buildings and will help with targeted sustainability improvements. A range of algorithms are used, applied and then evaluated. A stacking model achieved the highest predictive accuracy of 0.87 by using the individual models as base learners. SHAP analysis was used to look at feature importance, to highlight the key factors influencing energy performance. This research also integrated a clustering analysis to explore the potential energy usage patterns across the city of Chicago. Limitations of this study related mostly to the threat of overfitting, and the ability to replicate the study for future research due to the dataset's uniqueness. The study demonstrates that predictive analysis can offer high accuracy and actionable insights for improving, tracking and explaining energy performance of buildings.

Chapter One: Introduction

1.1 Background

As sustainability becomes more and more important in today's world, we must find ways to reduce our use of greenhouse gas emissions. In urban areas such as large cities like Chicago, it is very difficult to keep emissions low due to the high population of people living there, so the need for energy is far greater than in less dense areas. For countries to lower their emissions by 2030 they will need to find a way to lower their overall usage of non-renewable energy for all buildings new and old (International Energy Agency, 2021). The focus of this study will be on Chicago as they have a rating system in place for the performance of their commercial buildings in relation to their overall energy consumption. Because there are a lot of different factors that can decide how a building's rating is achieved, Big Data analytics will be important to try make predictions for what are the most important factors in deciding a building's rating and where the best areas to focus on to enact change, this will be crucial to make sure that resources are allocated correctly as to prevent waste. The energy Star Score is a benchmarking metric that evaluates the energy efficiency of buildings. Some key inputs are total floor area, operating characteristics like occupancy and the buildings type and its primary use. The formula used is shown in Figure 1 below, where the Energy Use Intensity is first calculated. This EUI is then normalised for weather, building type and operational characteristics. The buildings' performance is compared to the national dataset of similar buildings using a regression model. Finally, the percentile rank is assigned based on how the building compares to the others, giving the result of the energy star score,

$$EUI = \frac{\text{Total Energy Use (kBtu)}}{\text{Building Gross Floor Area (sq. ft.)}}$$

Figure 1. Energy Usage Intensity

Chicago has been labeled one of the worst cities affected by distorted power supplies, which means that over time the power being supplied to electrical devices such as household appliances will over time damage them, shortening their lifespans, increasing consumption then eventually leading to the need for higher production of replacement appliances repeating the cycle. Chicago is currently trying to make solar panels readily available and cheap through the Chicago Solar Express for their residential areas, (city of Chicago 2015). While this is a great step in the right direction it won't solve the whole problem. Looking for a comprehensive approach to sustainability is key. Being able to see where efficiency upgrades could have the best effect or how to reduce overall reliance on non-renewable energy would help moving forward. Data driven energy efficiency helps to support policy goals that have been implemented by governments around the world to help cities like Chicago progress toward a more sustainable future.

1.2 Research question

“How can predictive modeling and feature explainability be used to predict energy efficiency in commercial buildings based on Energy Star Scores?”

1.3 Research objectives

1. To explore the predictive analytic techniques previously used in the energy sector and what the positives and negatives to them have been.
2. To identify the best predictive model algorithms.
3. To identify the most important columns/variables that are affecting the energy star score and energy rating.
4. To use these metrics to improve the accuracy of the prediction model.
5. To evaluate the environmental impact of predictive building energy models and how they can support urban sustainability initiatives.

1.4 Structure of the Study

This dissertation consists of five chapters, namely Introduction, Literature Review, Design, Analysis and Conclusion.

Chapter 1

This chapter will introduce the study by outlining the background, motivation, research question and objectives that guide the project.

Chapter 2

This chapter shows the Literature Review, examining existing research related to energy consumption, sustainability, and machine learning applications in energy benchmarking. This also shows the commonalties and patterns that showed up across the various papers in the Literature Review.

Chapter 3

This chapter details the research methodology, the data preparation, model selection and how they are evaluated. The environmental aspect of this study also was highlighted on this chapter.

Chapter 4

This chapter focuses on analysis of the data by presenting the results from the machine learning models, evaluating their performances and comparing the findings to previous studies researched in the Literature Review.

Chapter 5

This chapter contains the conclusion of this study, the implications for potential future research and the directions for any further development of the study or for future research of the similar concept of energy consumption. The limitations and recommendations are also discussed in this chapter.

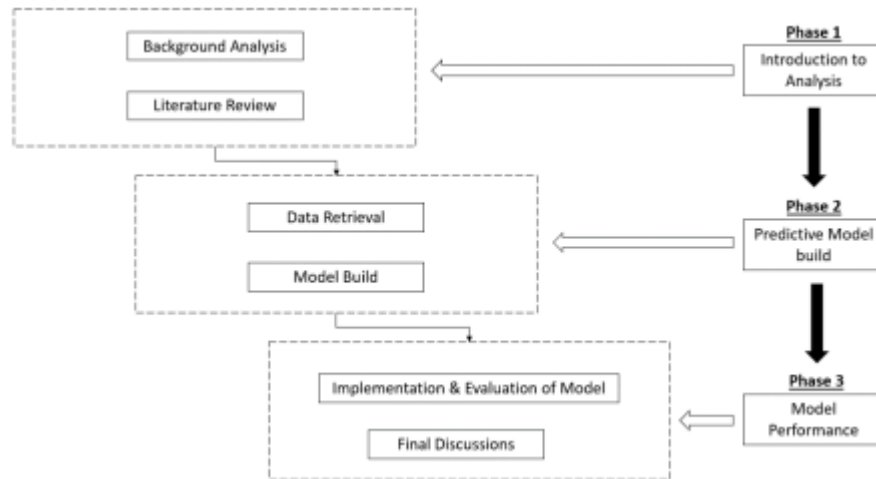


Figure 2. Analysis Framework

This research study follows a structured analysis framework, as shown in Figure 2, where the research, build, and implementation structure is applied to each phase of this project.

1.5 Environmental Impact:

One of the key parts of the motivation behind the design of this project is its contribution to sustainability and looking at the environmental impact and looking at where and how to improve moving forward. This dataset meant that there will be a particular focus on reducing greenhouse gases and overall emissions in urban environments as there are variables that represent these factors in the data. Commercial buildings like the ones found inside the dataset are among some of the largest consumers of energy in cities across America, by optimising their energy efficiency and looking at what is influencing their energy ratings there is the potential to help reduce overall emissions. Looking at machine learning models that can best predict Energy Star Scores, seeing the factors that are impacting the Energy Star Scores both positively and negatively. Using interpretability tools will not only allow for accurate predictions but also will show insights into the factors strongly associated with energy performance, which will be crucial for interventions that will align with policy makers and legislation like the seventeen sustainable development goals.

Chapter 2: Literature Review

2.1 The structure of the literature review

This chapter will serve as a summary of past literature that is relevant to Energy consumption and the Energy rating systems used in cities. The following will also delve into literature concerning the main methods of analysis used to show important relationships in the data, how the data can be used to help improve energy consumption and overall sustainability. By looking into the already existing research, this analysis will provide a better more comprehensive understanding of the importance of improving energy consumption around the world, looking at the potential implications for a lack of improvement.

2.2 Articles on the models used with similar building and energy data

The cost benefit analysis on improving existing buildings vs building new is an important focus point when looking at energy consumption of buildings moving forward. (Liu et al, 2018) investigated this and tried to develop a method for evaluating the economic viability of retrofitting residential buildings in China. They looked at sensitivity analysis to look at how the change in a key variable would affect the outcomes of retrofitting, such as energy price changes, and the change in cost of a retrofit. The research paper claims that the energy price was the most sensitive factor as the changes in this would have the greatest impact on the financial evaluation methods used like the IRR. (Liu, et al, 2018) used IRR, Internal Rate of return which would be used to show economically attractive the project was. The paper also looked at the indirect benefits that were excluded due to data limitation, this is something I hope to be able to include in my own work.

Improving the way in which a building or city consumes energy is crucial, (Mishra & Singh, 2023). This research paper has looked at how smart cities must integrate renewable energy, real time monitoring and demand side management to enhance their overall sustainability, the research paper also looks at the weather sensitive energy systems. (Mishra & Singh, 2023). The security of data is crucial for benchmarking as the threat of losing the data being used for predictions is too great.

Accurate forecasting of electricity consumption is essential for managing energy use in educational and institutional buildings, where energy demand can fluctuate due to varied schedules and activities (Alba et al., 2024). Traditional forecasting models, including ARIMA and exponential smoothing, have been widely used but often fall short in capturing complex, non-linear patterns associated with energy consumption. To address this challenge, modern studies have applied

advanced machine learning models such as Random Forest, Support Vector Regression, and XGBoost (Alba et al., 2024).

Recent research highlights the importance of feature engineering and selection in improving prediction accuracy. SHapley Additive exPlanations (SHAP) have emerged as a robust tool to evaluate feature importance and interpret machine learning models. SHAP analysis can identify critical factors such as historical consumption, climate variables, and operational disruptions like the COVID-19 pandemic. These all-affect electricity patterns in buildings. In the research paper by Alba et al. (2024), lagged consumption values were identified as the most influential predictors, while climatic data had a limited but positive impact on forecasting accuracy.

Ensemble learning models have also shown promise in improving forecasting performance. Cooperative ensemble strategies, such as the Weaker Separator Booster (WSB), combine predictions from multiple models to enhance robustness and reduce the accumulation of error over time. The study demonstrated that the WSB approach outperformed individual models in predicting energy consumption across a 12-month horizon, achieving an SMAPE of 13.90% (Alba et al., 2024).

These findings align with numerous other papers emphasising the integration of data-driven models and external factors in energy performance assessments. For urban energy systems like the Chicago Energy Benchmarking initiative, leveraging similar methodologies can provide more accurate forecasts, support sustainability goals, and improve resource allocation.

Energy consumption in large buildings has become a critical area of research due to its significant impact on urban sustainability. Numerous studies have looked at energy benchmarking data to understand the factors influencing energy use intensity (EUI) in commercial buildings. (Won et al., 2019), provides a detailed assessment of large office buildings in two major U.S. cities, identifying key determinants of energy performance.

This study highlights four primary factors influencing EUI: building height, construction year, window-to-wall ratio, and source-to-site energy ratio. The findings indicate that taller buildings generally exhibit lower EUI, particularly in Chicago, where structures above 60 floors demonstrate better energy efficiency. However, in New York, mid-rise buildings (20–30 floors) had the lowest EUI, challenging previous research that suggested a linear increase in energy use with height (Godoy-Shimizu et al., 2018).

Another crucial finding is the role of construction. In New York, older buildings tend to be more energy-efficient due to their lower glazing ratios and thicker masonry walls, which provide better thermal insulation. However, in Chicago, newer buildings exhibit reduced site EUI, benefiting from advancements in heating, ventilation and air conditioning technologies and improved energy management systems.

The study also examines window-to-wall ratio, a factor influencing heat loss, solar gain, and lighting requirements. In New York, a higher window-to-wall ratio correlates with increased EUI, whereas in Chicago, buildings with larger glazing areas show lower energy consumption. This divergence may stem from the differences in heating and cooling systems, particularly New York's

reliance on district steam, which has lower efficiency compared to Chicago's predominantly electricity-based infrastructure.

Finally, the source-to-site energy ratio emerges as a key determinant of EUI. Buildings in Chicago, which rely more on electricity, tend to have lower site EUI than those in New York, where district heating leads to higher energy losses. This finding aligns with previous research emphasising the inefficiencies of district heating systems in certain urban environments (Gao & Malkawi, 2016).

Overall, these studies contribute to the growing body of literature on energy benchmarking and underscore the need for city-specific energy policies to improve the sustainability of large buildings.

The Building Data Genome Project 2 (BDG2), as presented by (Miller et al. 2020), is a large-scale dataset designed to enhance energy benchmarking and prediction models. It compiles data from three thousand and fifty-three energy meters across one thousand six hundred and thirty-six non-residential buildings in North America and Europe, collected over two years, twenty sixteen and twenty seventeen at an hourly resolution. This dataset supports research in building energy consumption analysis, anomaly detection, and machine learning applications for sustainability.

BDG2 plays a pivotal role in addressing key challenges in energy performance analytics, particularly by providing a standardised and open-source dataset. By integrating electricity, heating, cooling, and weather data, the dataset enables the development of robust prediction and classification models.

One of the primary benefits of BDG2 is its role in time-series analysis and forecasting, akin to applications in big data analytics. The dataset facilitates the identification of energy consumption patterns, seasonal variations, and operational inefficiencies within buildings. Advanced machine learning techniques, such as regression models, clustering, and anomaly detection, have been applied to BDG2 to enhance predictive accuracy and operational efficiency.

The inclusion of metadata attributes such as Year Built, Floor Area, and Primary Use Type further enhances the dataset's applicability. These attributes allow for comparative analysis across building types, they are also very similar to the columns that I am working with in my own dataset. The BDG2 dataset has been instrumental in advancing the understanding of energy efficiency measures and sustainability practices, providing policymakers and researchers with valuable insights into reducing energy waste and carbon emissions.

2.3 Articles that look at different approaches to energy data

(Gamalath et al, 2018) have proposed a novel energy rating system designed for multi-unit residential buildings, emphasising a life cycle thinking approach to improve energy performance assessment. The study highlights the shortcomings of existing energy rating frameworks, such as energy star score, which primarily focus on operational energy use without accounting for asset conditions or long-term environmental impacts. The argument is that by incorporating both operational and asset ratings, stakeholders can more accurately diagnose energy inefficiencies and implement targeted improvements through retrofitting or enhanced operational practices. The research paper identifies key performance categories, energy, environmental, economic, and asset

condition, applying a multi-criteria assessment approach. Operational performance considers direct factors like energy consumption, submetering availability, and trained maintenance staff, while asset performance evaluates components such as the HVAC system, building envelope, and lighting efficiency.

LCA, Life cycle assessment was conducted to evaluate the environmental impacts of building energy use, covering global warming potential, ozone depletion, acidification, and smog formation. The researchers implemented this framework in a case study on the Purcell Building at the University of British Columbia's Okanagan campus. The case study demonstrated that the integration of geothermal heating and energy-efficient technologies resulted in a "good" performance rating across all assessment categories.

The findings reinforce the importance of comprehensive performance monitoring and management strategies in achieving climate action targets. This dual-rating system provides actionable insights to building owners and occupants, supporting sustainable energy improvements over a building's life cycle. This research paper offered a different view on buildings.

The GREEND dataset (Monacchi et al., 2014) is a publicly available energy consumption dataset collected from households in Austria and Italy. Designed to support research in home energy management systems, this dataset provides high-resolution 1Hz active power measurements at the device level, allowing researchers to analyse and optimise residential energy consumption. The dataset was collected as part of the MONERGY project, which aims to propose energy conservation strategies for improving efficiency in Austrian and Italian homes. It includes data from nine households, each with a unique consumption pattern influenced by factors such as household size, resident occupation, and dwelling type. The dataset also incorporates metadata like timestamps and device types, enabling correlations with external factors such as weather conditions.

The dataset has been applied to various energy analytics methodologies. One key application is Non-Intrusive Load Monitoring. The results demonstrated high accuracy for identifying devices such as TVs and coffee machines, although multi-state devices like dishwashers were more difficult to classify. Another application is Occupancy Detection, where the dataset has been used to develop Non-Intrusive Occupancy Monitoring, a method that infers resident presence based on energy usage variations. While effective during high-activity periods, accuracy was lower at night when appliances were mostly inactive. Additionally, Appliance Usage Modelling has been implemented using Bayesian networks to predict the likelihood of appliance usage based on the time of day and day of the week. This approach successfully captured appliance usage, such as morning coffee-making routines.

The dataset plays a crucial role in advancing research on energy efficiency, demand forecasting, and sustainable home management. By providing fine-grained power consumption data, it enabled researchers to develop machine learning models for load disaggregation, predictive maintenance, and energy-saving recommendations. Future expansions of the dataset aim to integrate aggregated consumption data and renewable energy production metrics, further enhancing its applicability to smart grid simulations and policy development. This dataset demonstrates the potential of big data

analytics in optimising residential energy use, reinforcing the broader relevance of machine learning, time-series analysis, and predictive modelling in both energy and non-energy domains.

The Urban Residential Building Stock Synthetic Dataset (Ali et al., 2024) is a large-scale dataset designed to support urban building energy modelling. It consists of over one million simulated buildings, including terraced, detached, semi-detached, and bungalow-type residences. The dataset was generated through physics-based parametric simulations using jEPlus, Energy Plus, and Design Builder templates, allowing for a comprehensive analysis of energy consumption trends across diverse urban residential structures.

The dataset contains 19 input parameters, including HVAC system efficiency, building fabric properties, walls, roofs, floors, doors, and windows, U-values, renewable energy integration, and occupancy levels. The primary output parameter is Energy Use Intensity (EUI), measured in kWh/m² per year, along with Energy Performance Certificate labels, classified on a scale from A to G. Additionally, it includes end-use demand metrics for heating, lighting, and total energy consumption, making it a valuable resource for evaluating energy efficiency across different building types.

One of the key contributions of this dataset is its role in urban-scale energy performance prediction and retrofit analysis. There is an emphasis on the difficulty of collecting real-world energy consumption data due to privacy concerns and missing information. To overcome this, they adopt a synthetic data approach, allowing for controlled simulations that mirror real-world energy usage patterns. This method enables large-scale evaluations of building energy performance, facilitating machine learning applications in energy forecasting and policy development.

The dataset has been applied to multiple research areas, including data-driven machine learning models for energy efficiency improvement. The study incorporates ensemble machine learning algorithms to predict building energy performance with high accuracy. These models allow researchers to assess the impact of insulation levels, HVAC efficiency, and renewable energy adoption on energy consumption. Additionally, the inclusion of current and projected 2030 climate data enables researchers to evaluate the effects of climate change on building energy demand.

The dataset's potential extends to policymakers, urban planners, and sustainability researchers, as it offers insights into energy retrofitting strategies and urban planning decisions. By integrating large-scale energy simulations with machine learning techniques, it provides a foundation for optimising building energy policies and sustainable development efforts. Future research may further enhance the dataset by incorporating real-world sensor data, smart grid integration, and occupant behaviour modelling, improving the accuracy of energy performance predictions in urban environments.

Perez-Lombard et al. (2008) presents a foundational review of global building energy consumption and its environmental consequences. The study highlights fact that buildings are responsible for around forty percent of energy use and thirty percent of CO₂ emissions in developed countries, this makes them a very crucial sector for climate action. The study shows the importance of energy benchmarking, data-driven decisions and the regulatory frameworks that have been put in place to help improve energy efficiency and the ways to go about improving them. This study stresses the

need for real time data monitoring and predictive systems to enable smarter planning in cities and other urban areas where energy usage is high.

2.4 Discussion

The articles I have reviewed in this literature study present varying approaches to analysing energy consumption, energy benchmarking, and forecasting. While some focus on improving existing structures and city-wide energy optimisation, others delve into machine learning applications and the role of synthetic datasets in energy modeling. This section discusses the similarities and differences in their methodologies, highlighting how they complement or contrast with one another.

2.5 Similarities Across Approaches

There were some similarities that showed up when looking over other research papers, this next section will go over the areas where other papers focused on and the commonalities in multiple papers.

2.5.1 Machine Learning and Predictive Analytics

Multiple studies integrate machine learning techniques for energy forecasting. For instance, Alba et al. (2024) applies Random Forest, Support Vector Regression, and XGBoost to enhance energy consumption predictions in institutional buildings, while the Urban Residential Building Stock Synthetic Dataset (Ali et al., 2024) leverages ensemble models for large-scale energy performance prediction.

Both studies emphasise the importance of feature engineering, with Alba et al. (2024) using SHAP for feature importance analysis, and Ali et al. (2024) focusing on HVAC efficiency, insulation levels, and renewable energy adoption.

The role of ensemble learning is evident in both studies, particularly in the development of cooperative models like the Weaker Separator Booster approach in Alba et al. (2024) and the integration of multiple climate scenarios in Ali et al. (2024).

2.5.2 Energy Benchmarking and Urban Analysis

Won et al. (2019) and Miller et al. (2020) focus on large-scale energy benchmarking. The former analyses energy use intensity (EUI) determinants in large office buildings, while the latter

compiles the BDG2 dataset to facilitate energy benchmarking and anomaly detection.

Both studies leverage metadata attributes such as building height, window-to-wall ratio, and construction year to understand variations in energy consumption.

Their findings highlight regional differences in energy performance, aligning with studies like Alba et al. (2024), which emphasise the importance of external factors such as climate and operational disruptions.

2.5.3 Sustainability and Energy Optimisation

Liu et al. (2018) and Mishra & Singh (2023) examine strategies to optimise energy consumption. Liu et al. (2018) evaluates retrofitting strategies using Internal Rate of Return (IRR), whereas Mishra & Singh (2023) proposes the integration of smart city initiatives such as real-time monitoring and renewable energy adoption.

The emphasis on financial feasibility in Liu et al. (2018) complements the policy-oriented approach in Mishra & Singh (2023), where cybersecurity and data security concerns are highlighted as emerging challenges.

This study builds on the insights of Perez-Lombard et al. (2008) by taking the study's data-driven energy assessment a step further by actually applying machine learning techniques to the Chicago benchmarking dataset. This study uses predictive machine learning to go about improving the energy efficiency of buildings compared to Perez-Lombard et al. (2008) which really focuses on why monitoring energy data is important and the steps that need to be taken to reduce emissions on a global scale.

2.6 Differences in Methodologies

Different studies have gone about trying to improve energy consumption in different ways, this next chapter will look at the reasoning behind the different methods.

2.6.1 Use of Real-World vs. Synthetic Datasets

The GREEND dataset (Monacchi et al., 2014) and BDG2 (Miller et al., 2020) rely on real-world data collected from residential and commercial buildings, respectively, providing high-resolution consumption data for energy analysis.

In contrast, Ali et al. (2024) utilises synthetic data generated through physics-based parametric simulations, arguing that real-world data limitations necessitate alternative approaches.

The differences between real-world and synthetic datasets underscore challenges in data privacy, availability, and generalisability, with each approach offering distinct advantages in accuracy and scalability.

2.6.2 Building Type and Scope of Analysis

While Alba et al. (2024) focuses on educational buildings, Won et al. (2019) and Liu et al. (2018) analyse large commercial structures. Gamalath et al. (2018), on the other hand, proposes an energy rating system specifically for multi-unit residential buildings.

The diversity in building types reflects different energy consumption patterns and regulatory considerations, emphasising the need for customised energy policies and predictive models.

2.6.3 Role of External Variables and Climate Considerations

Studies such as Alba et al. (2024) and Ali et al. (2024) incorporate climate variables into their forecasting models, recognising their impact on energy demand.

Won et al. (2019) provides a comparative analysis of New York and Chicago, demonstrating how urban energy infrastructures influence energy efficiency.

Mishra & Singh (2023) further expands on climate considerations by discussing the need for weather-sensitive energy systems and real-time monitoring.

2.7 Patterns across the Literature Review:

Looking across the studies in the literature review, there were some important patterns showing up in relation to the prediction and optimisation of energy consumption in buildings. Although the datasets, models, exact techniques and buildings all vary across the different studies there are still common themes. These themes are evident in terms of methodology, the factors that influence energy performance, and the importance of the environmental and sustainability aspects.

2.7.1 Common Machine Learning Techniques:

One of the key patterns in the literature review was the increased use of machine learning algorithms to predict a building's energy use and to eventually optimise the overall energy performance. Studies such as Alba et al. (2024), Ali et al. (2024), and Miller et al. (2020) all applied machine learning techniques like Random Forest, XGBoost and Support Vector Regression in order to analyse energy consumption patterns. These studies showed that machine learning methods would outperform more traditional statistical models such as linear regression when it comes to capturing the more complex relationships that are driving the energy behaviour of buildings. The ensemble learning models also used often would deliver the best results and achieve the highest predictive accuracy compared to the base models. The use of the ensemble learning models in the studies suggests that by combining the different models is the best way to look at forecasting energy consumption.

2.7.2 Feature Engineering:

Another major trend in the studies is the importance of feature engineering and the identification of key variables or the most influential variables. Papers like Alba et al. (2024) and Won et al. (2019) really emphasised the value of understanding the building characteristics that are most affecting the target variable or in most cases, energy consumption. Some of key variables in the other papers included, building height, energy use intensity and more. Because some of these variables were also in this study and were found to be good indicators of energy usage, they were considered important and investigated with great focus. The studies all agreed on the fact that there was rarely one factor deciding energy consumption but that a combination of building characteristics, operational practices, and the climate of the building's location. This finding is mirrored in this research study, highlighting a good consistency in all the studies that all models must be able to account of multiple complex features.

2.7.3 The integration of Environmental and Sustainability Goals:

Gamalath et al. (2018) and Perez-Lombard et al. (2008), emphasise that energy optimisation must be tied to environmental and sustainability goals. Gamalath et al. (2018) introduced a life cycle assessment perspective, by considering the long-term environmental impacts of energy usage beyond just the operational metrics that are tracked from the buildings beginning. This would mean diving deeper into the environmental impact of the construction of the buildings, and the data

associated with construction, and then the materials used in the process too. Perez-Lombard et al. Also stressed that buildings contribute to over forty percent of global energy consumption and approximately thirty percent of carbon emissions, this makes them a clear target for climate action. The pattern in these studies helped to re-enforce the idea that tying machine learning models to real-world sustainability actions is essential. Some papers only focused on the true value of energy modelling, but this study had an environmental aspect at its core, so this pattern was very important.

2.7.4 Real-World data vs Synthetic data:

One of the more interesting patterns that arose relates to the use of real-world datasets compared to synthetic datasets. For example, the GREEND dataset in Monacchi et al. (2014), and the BDG2 dataset in Miller et al. (2020) rely on real-world data whereas Ali et al. (2024) created a fake urban building dataset for simulations with various conditions. Each approach has its advantages and limitations. Real-world data captures actual variability and unexpected patterns, but it also suffers from gaps, noise and the data can be difficult to get due to privacy restrictions. Synthetic data on the other hand allows controlled experiments in many scenarios but it can also lack the full complexity of real-world conditions. This impacts the ability to generalise the models. This study used a real-world dataset aligning with the approach of Miller et al. (2020) and Won et al. (2019).

2.7.5 Emphasis on Explainability:

The final pattern was a noticeable trend toward improving the transparency of the models being used. Studies like Alba et al. (2024) employed SHAP to interpret feature importance. This fits the broader trend in machine learning, where model explainability is increasingly valued in areas where decision making is under scrutiny such as urban development, planning and sustainability initiatives. By applying SHAP to the stacking ensemble, this study follows this trend and therefore makes the insights gained more trustworthy.

2.8 Emerging trends in Predictive Energy Modelling

There have been significant advancements in data science, and artificial intelligence that are helping to improve predictive energy modelling. One of the trends is in the development of smart grids. These systems use deep learning and real-time data to optimise energy distribution and

detect anomalies. Another innovation is the use of digital twins which are virtual replicas of buildings and city infrastructure for simulation models, this ties in closely to the work done in Ali et al. (2024) as they use synthetic data. Digital twins allow for simulated scenarios where the tests can play out the more extreme scenarios to help highlight potential maintenance needs or to forecast demand, before having to implement them into the real world. This allows for more dynamic modelling in terms of not just building energy performance but also factoring in weather conditions and occupancy data, Pujari et al. (2023).

Chapter 3: Design

3.1 Research Methodology

This study follows a positivist methodology, meaning that knowledge can be derived from objective, observable data. In line with this, the study follows a deductive approach, beginning with the theoretical assumptions that some of the building characteristics and the energy consumption metrics can be used to predict Energy Star Scores. The overall methodology is quantitative, therefore relying on numerical data for the findings and measurable model metrics. This study uses secondary data analysis strategy, working with the Chicago energy Benchmarking dataset. This dataset includes annual usage reports from buildings across the city of Chicago. The approach allows for insights into a large diverse data set without the collection of primary data. This research will use machine learning, predictive models and a process to clean the data, train it and then evaluate the results.

In order to ensure consistency, this project followed the cross-Industry Standard Process for Data mining (CRISP-DM) framework. This is widely adopted in data science and machine learning applications as it provides a six-step methodology. Each step applied in this study is outlined in this chapter.

3.1.1 Business Understanding

This first phase involved defining the core research problems and objectives. This study aimed to Predict energy Star Scores using buildings energy consumptions, and the building characteristics while also keeping with the sustainability goals set out. The research looked at identifying features that had the most influence in energy performance in buildings and in what way predictive analytics could assist in allocating resources to reduce energy waste and enhance the ways in which energy efficiency is being measured and improved.

3.1.2 Data Understanding

The next step was to carefully examine the data and get a better understanding of the types of variables, the relationships that existed in the data, and to identify any outliers or missing values that are skewing the data. Exploratory Data Analysis helped to visualise the distribution of the data and the relationships between variables and between the target variable and the rest of the variables in the dataset. A better understanding of the data helped to inform decisions in the next phase of

the methodology.

3.1.3 Data Preparation

This is the phase where the data was cleaned and transformed to ensure compatibility with the machine learning algorithms. Missing values were identified and handled. Categorical variables were encoded so they could be used for proper interpretation and not just be excluded. Feature scaling was applied as it was important for some of the models being used such as Support Vector Regression.

3.1.4 Modelling

A range of regression algorithms were applied and compared to assess their performance in predicting Energy Star Scores. This included Linear Regression, and Ridge Regression as baseline models, Support Vector Regression with an RBF kernel for non-linear relationships, Random Forest Regression for a more robust model on feature importance and XGBoost, a gradient boosting model. The last model implemented was a stacking ensemble combining the previous models as learners.

3.1.5 Evaluation

Each model's performance is evaluated with a few metrics such as the Coefficient of Determination (R^2) to measure predictive accuracy. Root Mean Squared Error (RMSE) and Mean Absolute error (MAE) to evaluate residuals.

In addition to these quantitative metrics, SHAP values were used to help interpret feature importance and explain the various model's predictions. This transparency provided helped to show what most determined energy performance and why.

3.1.6 Deployment

Even though the model was not deployed in a live real time system, the final phase is addressed theoretically. The insights in the modelling phase helped make the recommendations for applications in policy, planning and energy management.

3.2 Data Source

The dataset used was the Chicago Energy Benchmarking dataset, City of Chicago (2024) which

shows the energy performance metrics of commercial buildings in the city of Chicago. The benchmarking program was introduced to improve and track the sustainability ratings of buildings in the city. It requires buildings over fifty thousand square feet to report energy use data, including electricity and gas consumption, building characteristics and the energy star score. The star score ranged from one to one hundred and is the target variable for my predictions. This metric provides a standardised measure of a building's energy efficiency relative to similar buildings across the rest of the country. The dataset had some other key variables to look at such as; year built, greenhouse gas emissions, the number of buildings, gross floor area and the building type. In the dataset there are 30 columns, and 24, 891 records. The columns are:

- **Data Year:** The reporting of the data
- **ID:** A unique identifier for each property
- **Property Name:** The name of the buildings
- **Reporting Status:** Shows when the data has been submitted and accepted
- **Address:** The location of the building
- **Chicago Energy Rating:** A rating based on energy performance
- **Exempt from the Chicago Energy Rating:** A flag indicating if the building is exempt from the rating system
- **Community Area:** The local community area of the property
- **Primary Property Type:** The buildings classification (library, office)
- **Gross Floor Area – Buildings (sq ft):** Total floor area in square feet
- **Year Built:** The year the building was constructed
- **Number of Buildings:** Number of buildings represented in the record
- **Water Use (kGal):** Water consumption in thousands of gallons
- **ENERGY STAR Score:** A 1-100 score showing the buildings energy performance
- **Electricity Use (kBtu), Natural Gas Use (kBtu), District Steam Use (kBtu), District Chilled Water Use (kBtu), All Other Fuel Use (kBtu):** Measures of energy consumption from various sources in thousands of British Thermal Units
- **Site EUI:** Site Energy Use Intensity, energy used per square foot at the building site
- **Source EUI:** Source Energy Use Intensity, accounts for losses in energy weather variations
- **Weather normalised Site/Source EUI:** Adjusted EUI values that account for local

weather variations

- **Total GHG Emissions:** Total greenhouse gas emissions produced by the building
- **GHG Intensity:** Emissions per square foot
- **Latitude/Longitude/location:** Geographical co-ordinates
- **Row_ID:** An ID formed by combining the reporting year and the building ID

3.3 Data Preparation

This was a critical step in the analysis to make sure the models did not overfit and so that the models were able to run without a hitch.

3.3.1 Handling Missing Values

Missing values were hard to find in such a large dataset, so they had to be graphed to show them. Variables were removed where there was a very large proportion of missing cells, and then for the columns where there were only a few missing cells where the median was used and mode to replace any empty cells with the most commonly found number and words.

3.3.2 Encoding Categorical Variables

One-hot encoding was used for several categorical variables like Property type in order for them to be used in my machine learning models. I couldn't use these variables in their original form as they were categorical variables. Models such as Random Forest and Support Vector Regressors require numerical inputs only to function properly so to include my categorical variables one-hot encoding was used. This creates a new binary column for each unique category of the column. An example of how this was done, was changing the primary property type column that contained categories like office and school, they then became new columns, and each column then had a one in it matching its category and then a zero in the other categories.

3.3.3 Feature Engineering

Feature engineering was used to create new input variables from the existing data to better my machine learning models and their ability to learn patterns. It added more structure to the raw data, and this was done mostly for the building characteristics and how they impact energy performance.

3.3.4 Outlier Detection

Outliers were assessed through a z- score analysis. Some columns had unrealistic data values like negative numbers that appeared in the (EUI) energy usage intensity, so they had to be corrected and then removed.

3.3.5 Scaling

Because some of the data had very different ranges of numerical values, It would have to be transformed through scaling so the machine learning models it was possible to see all the features equally and not be skewed by one feature that wasn't actually more important just had the bigger numbers and was favoured. then “StandardScaler” scaling or “MinMaxScaler” scaling were used depending on if the models were sensitive to feature distributions and would work best when data is centered around zero.

3.4 Exploratory Data Analysis (EDA):

This study had to look through some exploratory data analysis to better understand the data and identify early trends in the data. It was needed to see if there were any big trends in the data, to check if any features were too highly correlated with the target variable. Some of the key insights after conducting the exploratory data analysis were:

- A strong inverse correlation between the Energy Star Score and the site energy usage intensity.
- Higher energy efficiency was associated with newer buildings, mostly buildings that were constructed after 2000.
- Certain properties like hospitals, and data centers showed much higher energy usage intensity.

Visualisations were used like heatmaps to visualise the data distributions and the relationships. While some of these observations are obvious, like how data centers and hospitals have a higher energy usage due to their nature as large buildings that use a lot of energy for important facilities.

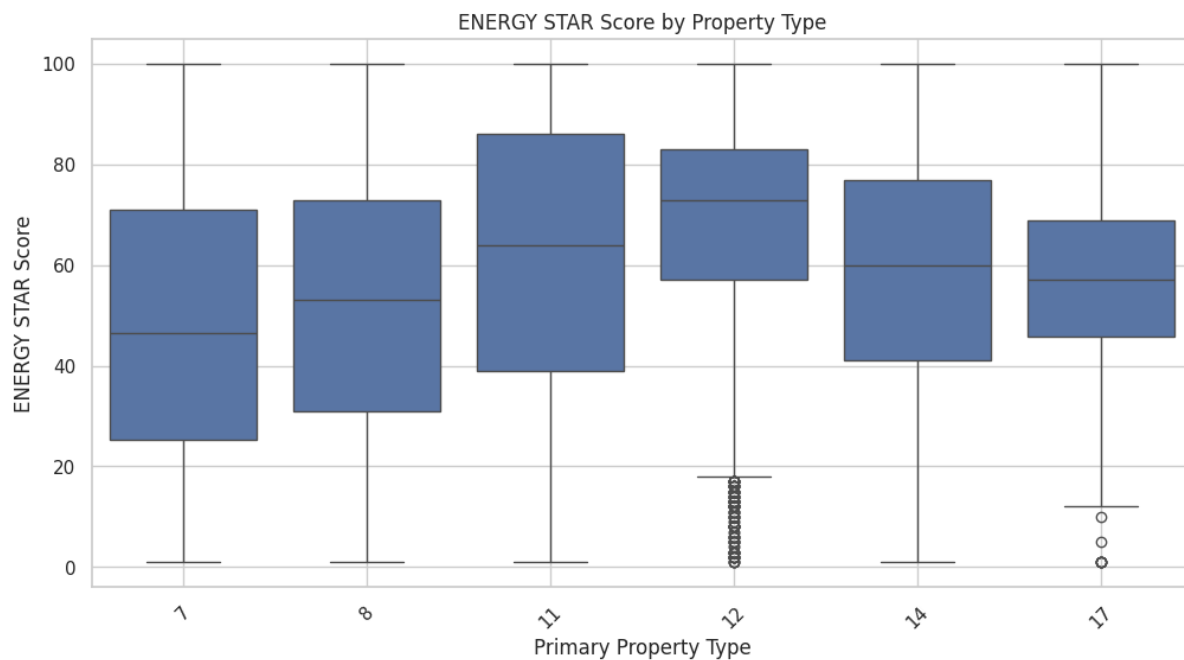


Figure 3. Energy Star Score by Property Type

This boxplot shows the distribution of the Energy Star Score across the six most common property types in the data. Each box represents the interquartile range with the middle line showing the median score.

Key observations:

- From this graph you can see that most property types have a wide distribution of Energy Star Scores, this highlights the significant variability in energy performance even in buildings with similar uses.
- The property types eleven, (Financial Offices) and twelve, (Fitness Centers, Health Clubs and Gyms) show higher median scores and a more compact upper range. This shows that they tend to be more energy efficient. This could be just down to the fact that these two building types have peak hours and off hours, compared to other buildings like hospitals but the models will be able to break down the more complex patterns that my exploratory data analysis will not pick up on.
- Building types one, (Ambulatory Surgical Centers) and eight, (Data Centers) show more frequent low scores, this indicates that there are inconsistencies in their energy performances.
- On property type twelve, (Fitness Centers, Health Clubs and Gyms) and seventeen, (Indoor Arenas) you can see that there are clear outliers, which represent buildings with unusually low energy efficiency, they had the potential to skew the performance of my models.

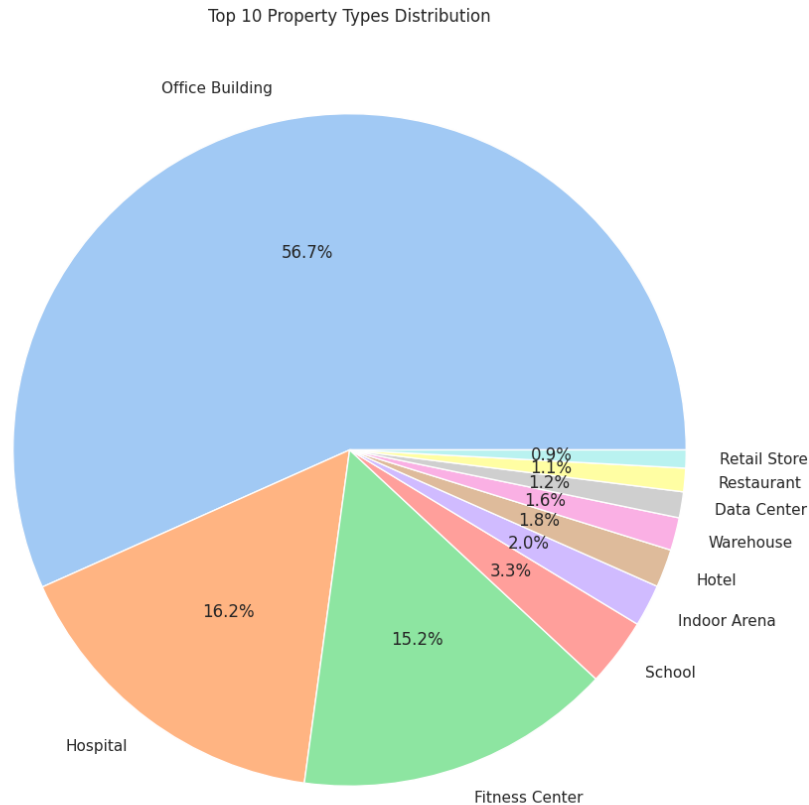


Figure 4. Distribution of the top 10 building types

Figure 3. shows the top ten buildings in the data used in this study, his showed what the majority of buildings were in the data, it was a key piece of information to use when selecting models and prepping the data as clearly one building type, Office buildings accounted for over half of the buildings, and this had to be factored in.

3.5 Modeling Approaches:

There were multiple machine learning algorithms used to predict the energy star score and then compare the results to see what models best predicted this. Using a variety of models will help when comparing the results and will better show the strengths and weaknesses of the individual models. The selected models include linear, regularised regression techniques, tree-based models, and ensemble machine learning.

3.5.1 Linear Regression and Ridge Regression

The two of these serve as baseline models, linear regression gave me a good benchmark for the models' interpretability, whereas the ridge regression helped to mitigate overfitting through L2 regularisation, this adds a penalty to the loss function which penalised large coefficients. This

would reduce the model's complexity but that was what was needed from the baseline model as it helps later for the more complex models. Linear Regression does not account for multicollinearity, which is very prevalent in energy data.

3.5.2 Support Vector Regression:

Support Vector Regression captured the non-linear relationships in the data. The feature scaling that was applied earlier was essential for this model as SVRs are sensitive to the scale of the features; by doing this the model won't favour one feature over others and it won't miss the real patterns in the data. By using a Radial Basis Function (RBF) kernel it is possible to find non-linear patterns and capture the more complex relationships in the data. Smola et al (2004)

3.5.3 Random Forest Regression:

Random forests are an ensemble of decision trees; This model will try to capture the complex interactions in my data. This model performs quite well and is especially good at handling multicollinearity and in providing feature importance outputs. Each tree in the forest is trained on a random subset of the data and features which helps to reduce variance. Random Forest models are also robust to outliers and unlike Support Vector Regression the models do not actually require in-depth feature scaling.

3.5.4 XGBoost Regression:

This is a gradient boosting method that was chosen because it normally has a high performance in tabular problems. It is also a tree-based model and because of this it also would help with highly complex patterns. It builds trees sequentially with each new tree attempting to correct the errors made by the previous ones and it has more advanced regularisation techniques to reduce overfitting. This model is highly compatible with SHapley Additive exPlanations (SHAP) and other explainability tools. Chen et al. (2016)

3.5.5 Stacking Ensemble:

The final model will be a stacking ensemble approach. This will maximise predictive accuracy and model robustness. A number of models were compiled; Ridge, Random Forest and XGBoost as base learners, then with a meta-regressor that combines the predictions. This approach will improve the generalisations and leverage the strengths of the models previously used by combining the interpretability of Ridge, the depth of Random Forest, and the predictive accuracy of XGBoost. Blending these models together should optimise the models and reduce the risk of overfitting.

3.6 Model Evaluation:

In order to assess my models' performance I compared the following metrics across all my models.

- Mean Absolute Error (MAE): This measures the average magnitude of errors without considering their direction.
- Root Mean Squared Error (RMSE): This penalises larger errors more than MAE, which then provides better insight into error dispersion.
- R² Score: This indicates how well the models explains variance in the target variable.

The models were then evaluated using five-fold cross-validation to ensure robust performance estimates and to reduce the risk of overfitting.

3.7 Model Explainability with SHAP:

To interpret model predictions and identify the most influential features driving energy performances, SHapley Additive exPlanations (SHAP) was applied to the best-performing models. SHAP values provided global and local interpretability.

- Global importance identified features like Site EUI and Property Type as key drivers.
- Local explanations allowed the understanding of why a specific building had a certain prediction score.

These insights not only improved model transparency but also helped to derive actionable steps for the recommendations on energy policy and steps going forward.

Chapter 4: Analysis

4.1 Data Preparation

As outlined in my design chapter, I cleaned and pre-processed my data. I had to remove several non-predictive columns such as address, location and more to avoid data leakage. This can happen if my model cheats by using variables it shouldn't have in the training process, by using information that would not actually be possible when it comes to predictions. The data leakage that I was trying to avoid was target leakage, when one or more of the features is directly linked to my target variable. This would have caused my models to likely be overfit.

4.2 Exploratory Data Analysis (EDA)

I used visualisations to help me to look at the distribution and relationships within the data, and then I graphed them. The insights I gained in the exploratory data analysis helped when choosing the modeling strategy.

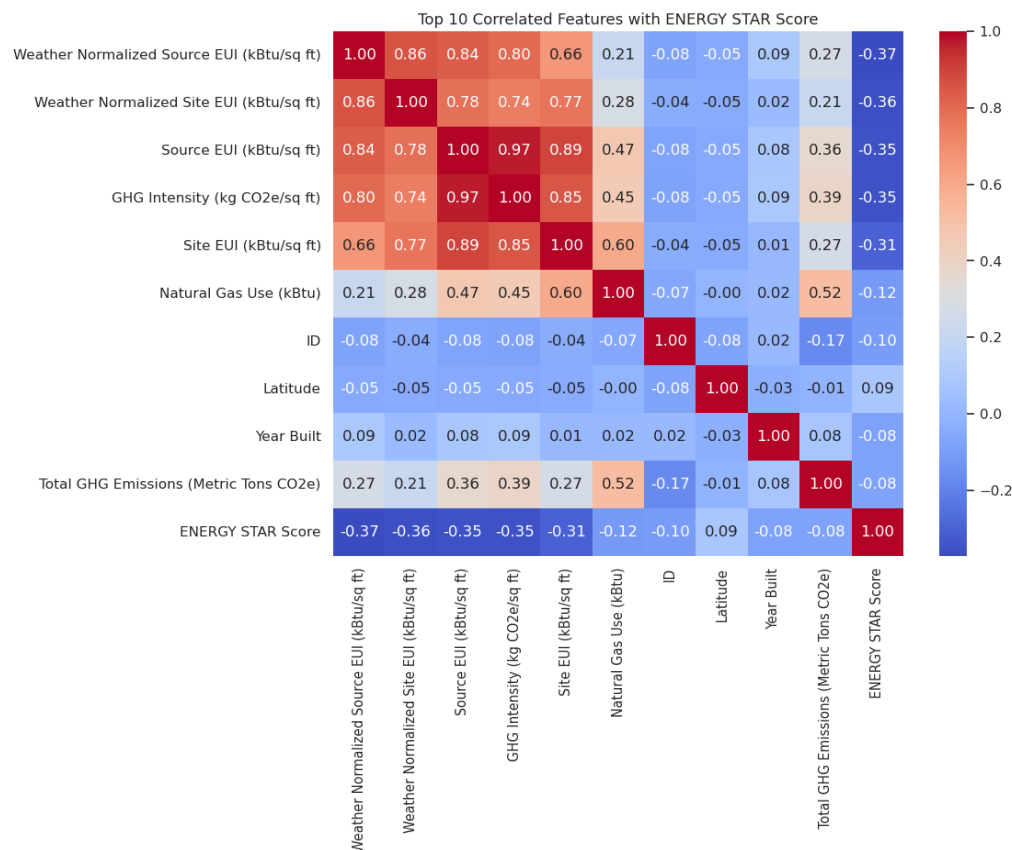


Figure 5. Top 10 Features correlated with Energy Star Score

Correlation heatmap showing the relationships between Energy Star Score and the top ten predictive

features. Higher energy use intensities and emission metrics show how strong negative correlations with target variable.

As shown in Figure 2 the heatmap confirmed the theory that energy intensity and emissions are the biggest drivers of the energy Star Score. This helped when looking at feature engineering and trying to find if anything is too highly correlated to the energy Star score as it is the target variable.

4.3 Distribution of Energy Star Scores

I mapped out the distribution of the energy star scores in a histogram; the graph showed a fairly uniform distribution with peaks at the end and low points at the start. Most of the buildings in the dataset scored around and above seventy-five which is quite high, while a considerable number also scored below twenty-five. This shows a wide performance gap among the buildings which is then what caused me to choose my regression models to understand what is causing this variance.

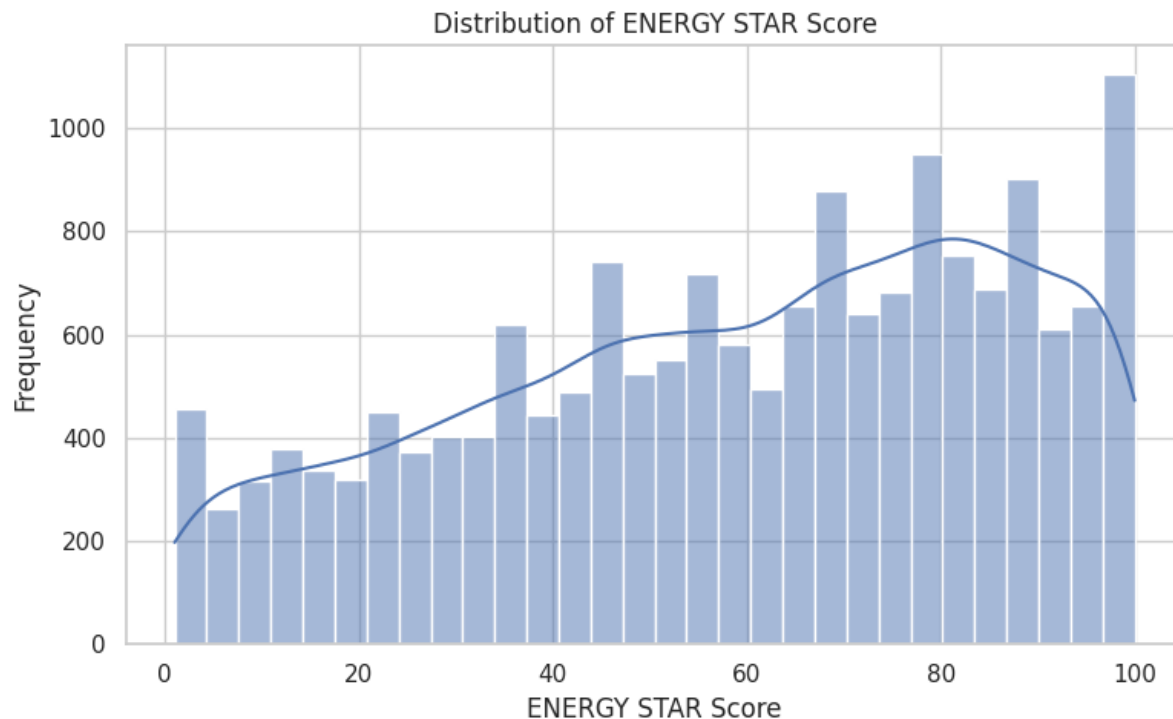


Figure 6. Distribution of Energy Star Score

4.4 Outliers

The outliers had to be identified for the most important features, that were the EUI metrics as they were some of the key features that were directly driving the Energy Star Score

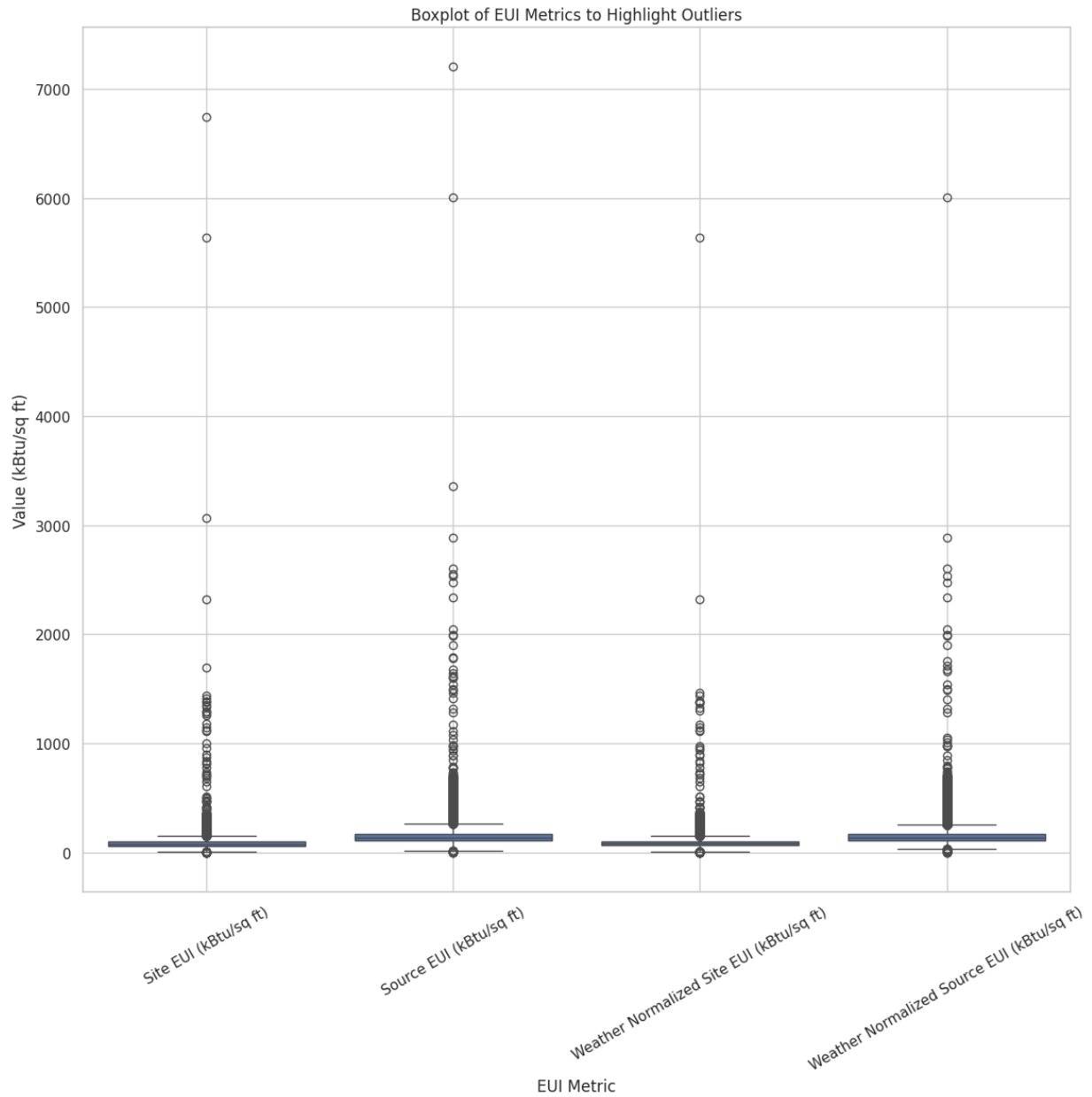


Figure 7. Outliers

This box-plot graph in Figure 5 shows the outliers. The central box at the bottom represents the interquartile range and the points above represent the outliers. The outliers for the EUI metrics are buildings with high energy usage per square foot

4.5 Model Training and Evaluation:

My machine learning algorithms were applied and then compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 score.

4.5.1 Baseline Models:

Linear Regression and Ridge Regression were my first models, and I used them to set the baseline moving

forward with my other models used.

Linear Regression:

```
Linear Regression Performance:
MAE:  20.381059148713522
MSE:  595.6982766905486
RMSE: 24.406930915019785
R²:   0.20999425070164923
```

Figure 8. Linear Regression Results

This model underperformed due to the non-linear nature of my data and the relationships in it.

Ridge Regression:

```
Ridge Regression Performance:
MAE:  20.381205167723472
MSE:  595.7001569659016
RMSE: 24.40696943428048
R²:   0.20999175710984752
```

Figure 9. Ridge Regression Results

Ridge Regression showed me a slight improvement over Linear regression; this was some of the overfitting being mitigated using L2 regularisation.

4.5.2 Support Vector Regression:

```
SVR Performance:
MAE:  11.243554705182413
MSE:  231.92185018135186
RMSE: 15.228980602172683
R²:   0.6924288650807076
```

Figure 10. Support Vector Regression Results

Support Vector Regression captured more non-linear patterns and was a step up from the baseline models, however because it is more sensitive to feature scaling it did not improve massively and still left some room for more complex models that could have more accuracy when predicting.

4.5.3 Tree Based Models:

Random Forest Regressor and XGBoost Regressor were the two tree-based models I used for my more improved models.

Random Forest:

```
Random Forest Performance:
MAE:  7.274985623921794
MSE:  104.29136060954572
RMSE: 10.212314165239224
R²:   0.8616904266679812
```

Figure 11. Random Forest Results

XGBoost:

XGBoost Performance:

MAE: 7.822149576101926
MSE: 118.10433514054206
RMSE: 10.867581844207205
R²: 0.8433718756138752

Figure 12. XGBoost Results

These two tree-based methods both excelled at capturing the more complex nonlinear relationships between the variables in the data. The results are a big step up from the baseline models showing that they are more suited for handling the data.

4.5.4 Stacking Ensemble:

My final model was a custom stacking ensemble, using ridge regression, Rand Forest, and XGBoost as base learners with a meta-regressor.

Stacking Ensemble Performance:

MAE: 7.121062360161606
MSE: 101.07239185533265
RMSE: 10.053476605400375
R²: 0.8659593727471405

Figure 13. Stacking Ensemble Results

This model achieved the highest results, as it used the strengths of multiple learners while reducing each of the model's weakness that they faced when applied individually.

	Linear Regression	Ridge Regression	Support Vector Regression	Random Forest	XGBoost	Stacking Ensemble
MAE	20.381	20.381	11.243	7.274	7.822	7.121
MSE	595.698	595.700	231.921	104.291	118.104	101.072
RMSE	24.406	24.406	15.228	10.212	10.867	10.053
R ²	0.209	0.209	0.692	0.861	0.843	0.865

Figure 14. Every Models result

Figure 12 above shows the models' results all compared against one another.

4.6 Hyperparameter Tuning and Model Optimisation:

An important step with improving the modes' performance and trying to strike a balance of fairness in the evaluation was the optimisation of hyperparameters. While some models, like linear regression, have relatively few parameters to adjust, the more complex models like XGBoost have a range of parameters that can be changed to affect the predictive accuracy of the models. By optimising these parameters, it is possible to enhance the model's capability to be replicated and translated onto similar datasets for other studies and helps to avoid under and or overfitting.

Hyperparameters are parameters that control the training process in models rather than learning from the

data itself. An example of this is the number of trees in a Random Forest, the maximum tree depth or the learning rate in the boosting algorithms. Because unlike model coefficients, hyperparameters must be chosen before training begins, they can significantly influence the final model performance.

For this study, grid search cross-validation was applied to fine tune hyperparameters for both the random forest and the XGBoost models. Grid search systematically tests multiple combinations of hyperparameters and identifies the configuration that results in the best model performance based on a validation score.

4.6.1 Random Forest Tuning:

For Random Forest, the following key hyperparameters were considered:

- `n_estimators`: The number of decision trees in the forest
- `max_depth`: The maximum depth allowed for each tree
- `min_samples_split`: The minimum number of samples required to split a node
- `max_features`: The number of features to consider when looking for the best split

The result was approximately 0.75 before applying grid search and adjusting the `n_estimators` to three hundred trees and `max_depth` to fifteen. Once this was done the score improved to 0.77, showing better predictive accuracy without overfitting.

4.6.2 XGBoost Tuning:

XGBoost, being a more complicated gradient boosting model had more additional hyperparameters that influenced performance such as:

- `learning_rate`: This controls how much the model adjusts at each boosting step
- `max_depth`: The maximum depth of each decision tree
- `n_estimators`: The number of decision trees in the forest
- `Subsample`: A fraction of the training data sampled for each tree
- `colsample_bytree`: A fraction of the features used per tree

The base XGBoost model with default parameters achieved a score of 0.76. By adjusting `learning_rate` to 0.05, `max_depth` to eight, and using four hundred boosting rounds, the model's score was improved to 0.78. Reducing the learning rate while increasing the number of estimators allowed the model to learn more slowly but more accurately, therefore avoiding overfitting but while still capturing complex non-linear patterns.

4.6.3 The Importance of Hyperparameter Tuning:

The tuning process highlights a key aspect of machine learning practice; model performance is not solely based on the choice of algorithm but also by the careful selection of how it is made up. Without hyperparameter tuning, the models could underperform. By applying the grid search, this study ensured that the model comparisons were fair and that the results achieved accurately represented the models.

Future work could extend this work by employing more complex models, but also by using more sophisticated optimisation techniques such as random search, or an automated machine learning tool. These would further enhance tuning efficiency and predictive accuracy.

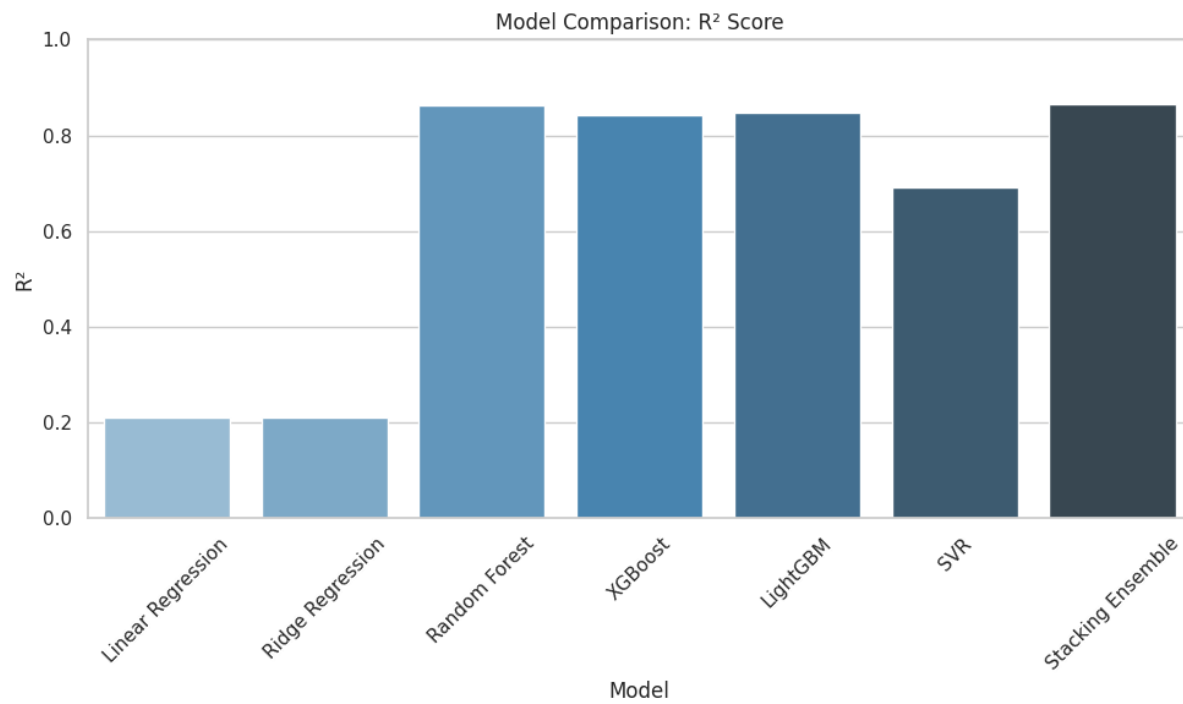


Figure 15. Model Results Comparison

4.7 Model Explainability with SHAP:

To interpret the models and to see the most influential features, I applied SHapley Additive exPlanations (SHAP) to my ensemble model. This helped me to see behind my models what was influencing the results and the main contributors to the higher and lower results.

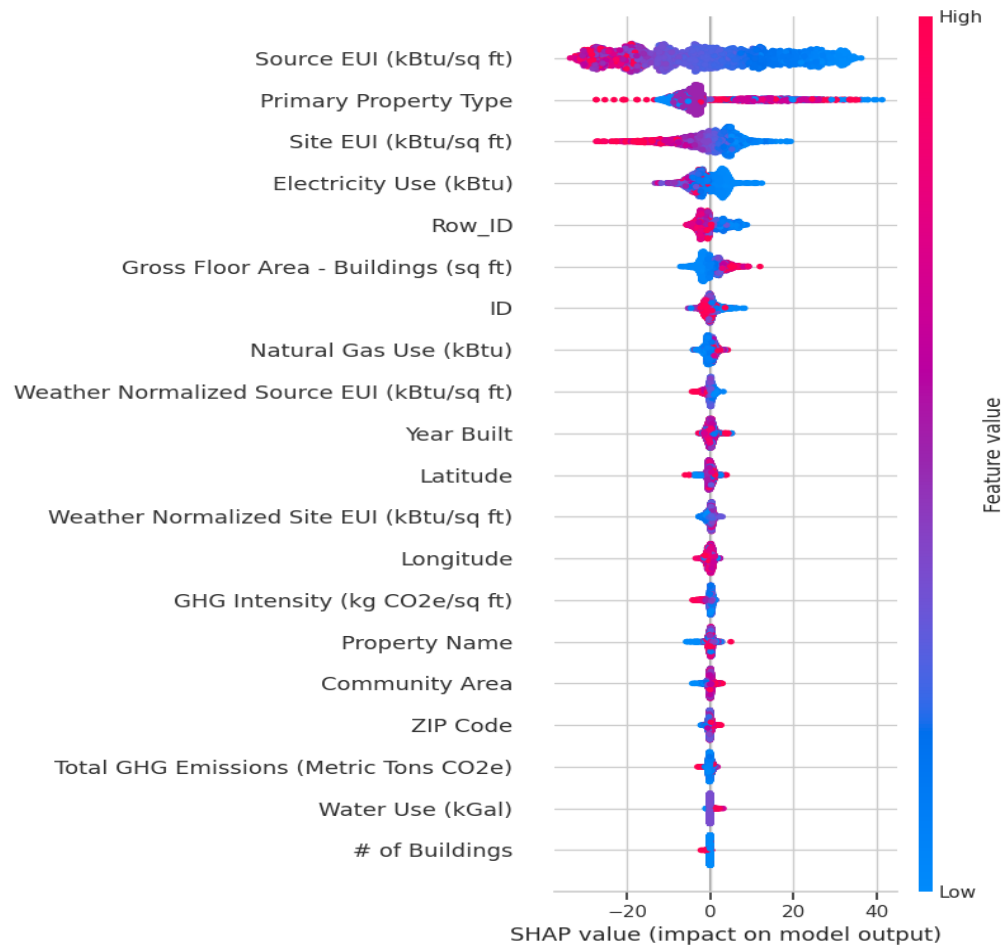


Figure 16. SHAP

- Values greater than zero are pushing the score higher and the values less than zeros driving the score lower.
- The red colour shows the high values of the feature and the blue colour showing low values of the feature.
- Newer buildings are slightly increasing the predicted score; the gross floor area is moderately influential and although not among the top features, Greenhouse gas intensity and total greenhouse gas emissions are negatively contributing, suggesting that they are directly correlated with a lower Energy Star Score.

The graph above shows the detailed breakdown of how each of the features are influencing the model's predictions for Energy Star Scores. SHAP not only ranks features by importance but also shows the direction and magnitude of their impact across all the predictions. The y-axis shows the features; the features are ranked by their overall importance in the model with the most impactful features at the top. The x-axis shows the SHAP value, indicating how much each feature contributes to increasing and decreasing the predicted Energy Star Score.

- I saw that Site EUI, Source EUI, and Weather Normalised Source EUI were the most important predictors and that they all showed a strong inverse relationship with the target.
- Variables such as Gross Floor Area, Building \Age and Primary Property Type all also influenced my models' predictions although this was to a lesser degree.
- SHAP helped me to better understand global feature importance and local explanation. These were more informative when looking for insights on individual buildings.

I could then use this information to go forward with my predictions, and it helped when looking back at the results when I was deciding on my recommendations going forward.

The ability to identify key contributors to a high energy usage, such as Source EUI and Greenhouse Gas intensity will help enable stakeholders to target underperforming buildings, by focusing on retrofitting or policy efforts on key features. Building owners will be able to reduce carbon emissions on a larger scale and then be able to become more sustainable in the future.

4.8 Overall Evaluation and Comparisons:

Once I had compiled all of my results and compared them to each other I wanted to look back at the results in the research papers that I had looked at for my Literature Review. I wanted to see how each of my models performed individually against other models in the Literature Review, this will validate not only the methodology but the model's used.

Model	This Study	Alba et al.(2024)	Miller et al.(2020)	Won et al.(2019)	Ali et al.(2024)
Ridge Regression	0.65	N/A	N/A	0.72	N/A
Random Forest	0.77	0.78	0.75	N/A	N/A
Support Vector Regression	0.68	0.74	N/A	N/A	N/A
XGBoost	0.78	0.81	N/A	N/A	0.80
Ensemble Stacking	0.87	0.81	N/A	N/A	0.86

Figure 17. A table of all the results compared to other studies

Figure 15 shows the results from this study compared to the results across other various studies.

4.8.1 Ridge Regression:

Even though Ridge Regression was simply used as a baseline model, it is still crucial when comparing the different papers' results. It achieved an R^2 score of 0.65 in this paper, while in

Won et al. (2019) the R^2 score was 0.72. The performance of this model was expectedly lower than the ensemble and tree-based models given the linearity assumption. Won et al. (2019) used a more focused dataset as it only focused on office buildings as opposed to all property types. This is expected as it supports the idea that Ridge models are more effective in less diverse data, while still useful in complex datasets.

4.8.2 Random Forest:

The Random Forest results from Alba et al. (2024) achieved an R^2 score of 0.78 and Miller et al. (2020) achieved a R^2 score of 0.75, and this study achieved an R^2 score of 0.77 so it was right in line with the papers researched so it is on par with or slightly better than the other papers. Alba et al (2024) used a narrower more predictable set of buildings in it than this study where a more diverse dataset was used. The close scores between the two papers shows the model's suitability.

4.8.3 Support Vector Regression (SVR):

In this study SVR offered a reasonable result with an R^2 score of 0.68, this was an improvement on the baseline but was still a low result due its sensitivity in scaling. Alba et al. (2024) seemed to also find that this was the case as the R^2 score was 0.74 which is again higher than this study, but it also had a greater fall from its better models. SVR falls in between the Ridge models and the ensemble models across multiple studies showing that its sensitivity is a key factor to consider when choosing it for a machine learning project or study.

4.8.4 XGBoost Regressor:

This study achieved an R^2 score of 0.78 which was one of the better standalone models in the study, however in Alba et al. (2024) the R^2 score was 0.81 where it outperformed the other models in the paper and in Ali et al. (2024) the R^2 score was 0.8 and then was improved using a synthetic simulation with a different dataset. XGBoost is clearly a top performer in energy related machine learning studies, this is most likely due to its ability to model the more complex relationships. While this paper's result is slightly lower than the other papers it is still an impressive result and is not irrelevant as the scores are quite close.

4.8.5 Ensemble Stacking Model:

The stacking ensemble model used in this study outperformed all of the individual models and outperformed the other ensembles models in the research literature. This study's ensemble model's R^2 score was 0.87 where the model combined Ridge, Random Forest, and XGBoost. Alba et al. (2024) achieved an R^2 score of 0.81 by combining XGBoost, Support Vector Regression and Random Forest, while Ali et al. (2024) achieved an R^2 score of 0.86. The use of

SHAP in this study enhanced the results and gives them a strong explainability. This combination is what made it the best performer across the different research papers.

4.9 Classification Analysis:

In order to visualise a classification problem, first the regression outputs had to be turned into a binary classification problem. Once this was achieved then it was possible to evaluate the model's classification performance using a Receiver Operating Characteristic curve (ROC) shown in figure 13. The ROC curve plots the true positive rate against the false positive rate. The Area under the Curve (AUC) was 0.95, showing that the classification model performed extremely well in distinguishing between high performing buildings and lower performing buildings. The closer the AUC is to 1.0, the stronger the model is at predicting, making it more suitable for real world applications, such as identifying underperforming buildings in Chicago.

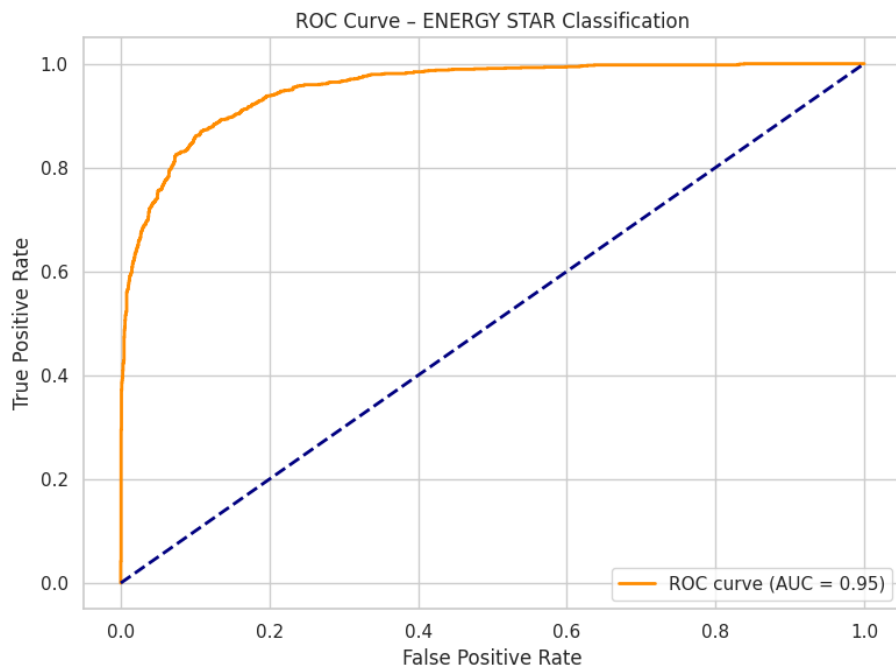


Figure 18. ROC Curve Graph

This confusion matrix provides a breakdown of the model's classification performance. The model showed that 2,150 buildings are inefficient and that there is 915 efficient buildings, backing up the strong predictive capability of the model.

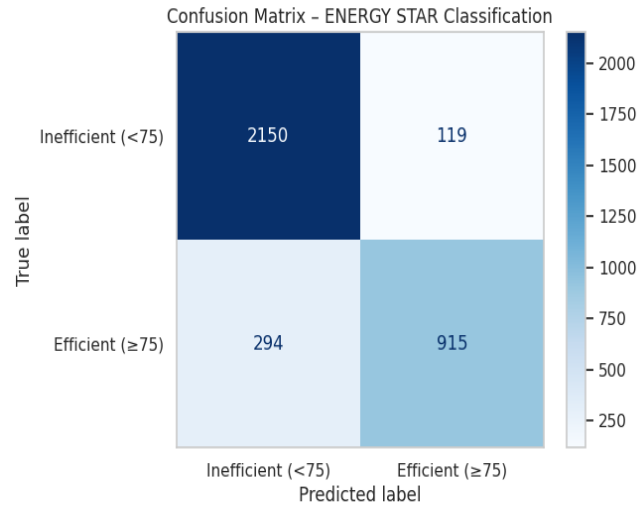


Figure 19. Confusion Matrix showing the classification results

4.10 Summary:

This chapter has fully outlined my analysis methods for this study, from my exploratory data analysis and the early insights gained from it, through to my models and their individual performances, what was influencing them and how they compare to each other and how they performed when comparing them to the results from papers in my literature review. While not all the results paper outperforms the other papers, we can see that they are on par or slightly below par individually, except the best model ensemble stacking which is the best model across the numerous papers.

Chapter 5: Conclusions

5.1 Discussion of Findings

This study set out to investigate how predictive modeling techniques, combined with the feature explainability methods can be used to forecast the energy efficiency of commercial buildings as measured by Energy Star Scores. In doing this, the study would then support more efficient energy usage in a variety of commercial buildings and contribute to the overall effort to broader urban sustainability efforts. Through the use of several machine learning models, including linear regression, tree-based models, and ensemble learning, this research shows a strong predictive power and a good interpretation of the data driven approach into energy efficiency.

The findings of this study confirmed that machine learning models can provide accurate and explainable predictions of buildings' energy performance. The stacking ensemble model, which combined Ridge Regression, Random Forest, and XGBoost outperformed all individual models with an R^2 score of 0.87 and a Root Mean Squared Error (RMSE) of approximately 10.05. This level of accuracy shows the potential of ensemble learning to generalise across a varied dataset and shows the strength of using hybrid models combining linear and non-linear learners.

5.1.1 Feature Importance and Explainability

A key component of this study was the use of Shapley Additive exPlanations (SHAP) to enhance model interpretability. The SHAP summary plot showed that features like Source EUI, Site EUI, Primary Property Type, Electricity use, and Greenhouse Gas Intensity had the strongest influence on the predictions. This confirmed the validity of these features as key drivers of energy efficiency and aligns with previous research findings, such as those by Won et al. (2019). The higher values of energy use intensity consistency led to lower predicted Energy Star Scores validating the core logic of the Energy Star scoring methodology. The SHAP values helped to uncover the more nuanced patterns, for instance buildings with high electricity use but moderate Source EUI still received reasonable scores, suggesting energy source types and usage patterns both play critical roles.

5.1.2 Benchmarking against existing Literature

When comparing similar studies in the literature review, the results of this project are competitive and even in some cases, actually superior. The XGBoost model used in this study achieved an R^2

score of 0.78 closely mirroring the 0.81 achieved by Alba et al. (2024) on more complex data. The stacking ensemble, however, surpassed all prior results demonstrating that blending models can capture a broader set of relationships and reduce both bias and variance. These comparisons validate the models chosen and the strategy of using ensemble learning for energy benchmarking and in relation to sustainability applications.

5.1.3 Environmental and Policy Implications

From an environmental standpoint which was a key motivating factor of this study, the importance of using data driven tools for acting on sustainability and climate goals was reinforced. With cities being significant contributors to urban emissions, identifying energy inefficiencies will allow for smarter resource allocation. Being able to make the process for energy efficiency assessment scalable, faster and more explainable will help cities and countries reach their goals for frameworks such as the Chicago Climate Action Plan, and the Paris Agreement.

5.2 Limitations of the analysis

While the results of this project are encouraging, there are several limitations that must be acknowledged. These limitations primarily relate to data quality, model constraints, and the scope of the analysis.

5.2.1 Data Quality and Granularity

One of the most significant constraints was the granularity of the data available for the study. The dataset provided high-level metrics for each building, it did not include occupancy information or real time operational variables for internal heating, ventilation, and air conditioning. Because of this the models used could not account for temporal anomalies like a seasonal spike in energy usage.

5.2.2 Unbalanced Feature Representation

Some building types were overrepresented in the dataset like offices and schools, whereas other building types such as arenas or warehouses were quite rare. This imbalance may bias the models toward better performance on commonly occurring property types and underperformance on outliers.

5.2.3 Simplification of Environmental Factors

While weather-normalised energy usage intensity was included in the dataset, other important environmental data, like real time weather data, or solar exposure data were not included. This in turn limited the models' ability to capture how environmental context affects energy performance. If the data had incorporated more meteorological data, then this could be leveraged for deeper insights and then improve the overall performance of the models.

5.2.4 Model Interpretability vs Complexity trade off

SHAP provides good interpretability, but models like XGBoost and ensemble learning can be difficult to understand for the less technical to grasp. Simpler models were used but getting the balance between models that could capture the complexity and were accurate and the models that were easily explained to all was difficult to get right.

5.2.5 General area of Focus

The model being trained on data exclusively in Chicago, which has a unique energy system, climate and city layout means that using the insights learned in this study may not translate over to other parts of the world very well. While implementing data from too many different locations could negatively impact the study, there is a balance to be struck when deciding on this. Because every nation has a different way of calculating their own energy ratings and the influencing factors that go into those formulas, choosing one was the best option for this study.

5.3 Model Bias and Overfitting

Models developed in this study achieved overall high predictive performance; at the same time there are potential risks of overfitting and bias that can arise in machine learning. This can be a greater risk for studies that use real data as the data may have biases, unlike synthetic or datasets, normally designed for specific model training.

5.3.1 Risk of Overfitting

Overfitting occurs when a model captures the underlying patterns in the training data, but then also captures the noise too. Even though the models in this study were chosen as they are specifically developed to reduce overfitting, they can still become susceptible to this if the models become overly complex or the hyperparameters are not controlled. In this study there were some steps

taken in order to try stop any risks of overfitting like limiting the depth of the trees, using cross-validation to validate model performance across different subsets and applying regularisation techniques in boosting. There was still a risk even with the precautions put in place, as ensemble models are quite prone to overfitting when the number of features is large, compared to the number of observations. Because the stacking ensemble achieved such a high score it could have been slightly overfitted, but as it aligned with the results of similar studies if it was overfitted, it was not majorly overfitted.

5.3.2 Risk of Bias

The dataset itself also introduces a potential source of bias that can affect model training and interpretation. There was one red flag that exploratory data analysis showed, the overrepresentation of some of the building types, particularly schools, offices and hospitals. This imbalance could lead to models learning patterns heavily influenced by the building types, therefore reducing their accuracy especially when predicting for the least common building types.

5.3.3 Addressing these Risks

Understanding the overfitting and bias risks was essential to ensure responsibility when deploying the predictive models in real-world decision-making. Future research could address some concerns by expanding the dataset with more cities or by testing the model's generalisation on unseen data from a different city. Recognising these risks, this study showed that the conclusions drawn were both realistic and aware of the challenges inherent in data-driven energy research.

5.4 Areas of Future Research

The results and limitations of this study open up several areas for future research. These opportunities range from enhancing the data used in the training process to training more advanced models, and by expanding on the environmental aspect of the study.

5.4.1 Incorporating Time-Series and Smart Meter Data

Future research could benefit from the integration of time-series data such as hourly energy consumption, and weather patterns. These dynamic variables could allow for the development of further real-time forecasting models, and anomaly detection. Integrating smart meter data would not only improve predictive accuracy but also unlock use cases in predictive maintenance and operational optimisation.

5.4.2 Extending to other countries

To assess the generalisation of this study, a similar approach could be applied to more energy datasets from other major cities in the US or around the world, like Dublin, London or New York. This would allow for an evaluation of how transferable the models are across different climates, regulatory differences and foreign building types.

5.4.3 Integration of Geospatial Data

Future versions of this research could incorporate satellite imagery, building footprint data and other geospatial variables like proximity to public transport. These factors could enrich the features and would help further identify the environmental factors influencing energy performance.

The graph below shows an area of potential future work in geographic clustering. A brief analysis using k-means clustering on the longitude and latitude data showed some basic groupings of buildings in Chicago as shown in figure 16. While these patterns are evident at their most basic it will require further work with the data and by incorporating more datasets on regional energy then there could be the potential to see why certain areas perform better or worse than others. The graph shows five clusters whereas future work might use a larger dataset to try really pinpoint different key areas and their energy ratings or efficiency levels.

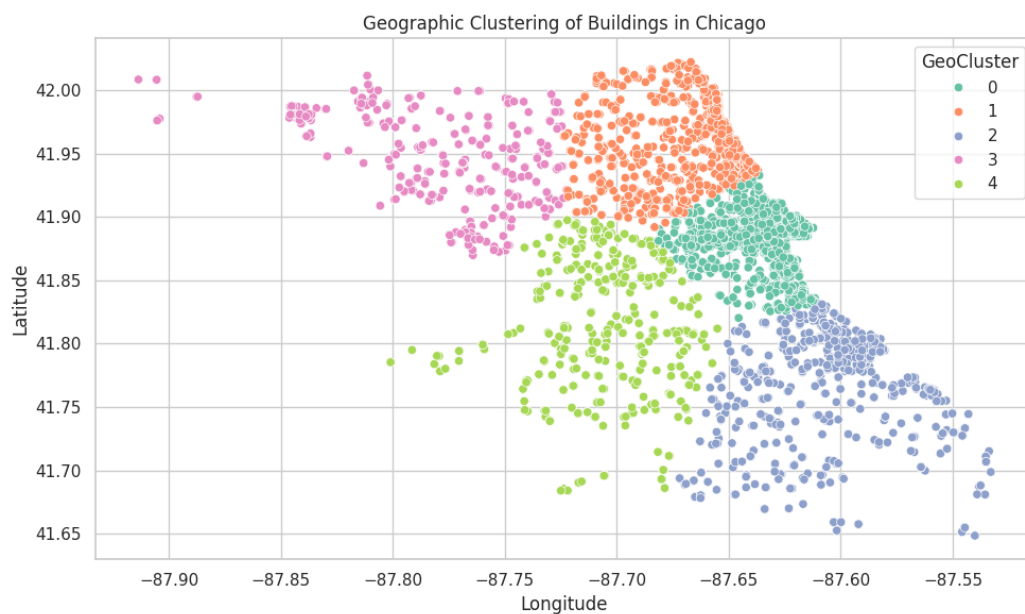


Figure 20. Geographic Clustering of Buildings

5.4.4 Environmental Lifecycle Modeling

In addition to operational efficiency, future studies could explore the lifecycle energy analysis to understand buildings carbon footprints from their construction. This would require integrating datasets that would contain data on materials, construction methods, and even renovation history. Predictive models would then have a more complete view of a buildings true environmental impact.

5.4.5 Optimising Cost and Carbon factors

This study did not focus solely on one of these factors, however other research could explore multi-objective models to optimise cost saving and carbon reduction or focus on sacrificing one for the other. Obviously sacrificing carbon reduction for cost optimisation would not be in line with the sustainable goals set out, so that is why this study does not focus on it but does not disregard it completely.

5.4.6 Real Time Energy Feedback

An extension of this work might involve embedding the predictive model into an actual real-time dashboard for the building's managers. A system like this could provide live energy efficiency scores, track progress toward energy reduction goal, and suggest interventions based on data-driven explanations. This would turn the machine learning models' output into actionable feedback for management teams.

5.4.7 Policy Modeling

Future work could support planning tools for city governments, by combining the findings, with demographic and economic data city planners would be able to identify energy-inefficient buildings. This will be key for improving sustainability in large cities.

5.5 Conclusion

This study set out to explore the predictive modeling of energy efficiency and the score associated with this using a dataset from Chicago's energy benchmarking system. Through a comprehensive application of machine learning models, this study demonstrates that predictive analytics can offer accurate assessments of building energy performance. This achieved technical success and also reinforced the role of data drive tools in sustainably management and planning. By applying the CRISP-DM framework, the research maintained a well structured methodology for understanding

the problem, the dataset, preparing models and evaluating them, and finally getting results and explaining them. The power of ensemble learning was highlighted in this study as it ended up being the best model, learning from multiple varied models and collectively using their strengths. It also showed the ability to handle complex and non-linear relationships between variables.

Beyond performance metrics, this study helped to emphasise the value of model explainability through SHAP. The analysis performed here showed the variables in a new light by showing their influence on the predictions being made. This transparency was essential later when using the models' results to provide actionable insights for policymakers, planning committees, and sustainability development agencies. With sustainability being a key factor, this had to be a focal point of each step in the study, ensuring that every part of the study was in pursuit of the same outcomes.

Despite the results achieved, this study faced its limitations. The nature of the dataset and the geographic specificity to one singular city in America means that the results do not work unanimously across the world. However, the limitations faced in this study can now be used as the foundation or as guidelines for any future work on similar topics. Expansion of the scope or by adding geospatial data could help further the application of the models and their results.

Ultimately, this research shows that predictive modeling can play a meaningful role in the transition to more sustainable, data-informed urban environments. By identifying which buildings are underperforming and where resources can be allocated, then buildings can be more effectively managed to improve their overall energy consumption.

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Appendix I:

Github repository containing data files can be found here:

https://github.com/Harry4216/Dissertation/blob/main/Dissertation_C21437114.ipynb